Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.C.A DEGREE EXAMINATION APRIL 2021

CA102T - DIGITAL LOGIC FUNDAMENTALS

Time : 3 hrs **Max Marks :75**

SECTION - A (5X5=25)

Answer any **FIVE** Questions

1. Convert the following decimal numbers to binary and its octal equivalent.

   a) $(174)_2$ a) $(256)_2$

2. What is an universal gate? Explain.
3. Describe Quine Mcclausky tabulation method.
4. Give a short account on Programmable logic array.
5. Explain the significance of Demultiplexers.
6. Describe the working of 3-8 line Decoder.
7. Write a brief note on the working of RS Flip Flop.
8. Explain ripple counters.

SECTION - B (5X10=50)

Answer any **FIVE** Questions

9. Explain the basic logic gates with its truth table.

10. Describe the Karnaugh map method with an example.

11. Explain the various laws of Boolean algebra.

12. Illustrate the working of Full adder and Subtractor with a neat diagram.

13. Simplify the following using K-map method and draw the logic circuit
    $F(A,B,C,D)= \sum (0,2,3,5,8,9,10,11,13,15)$

14. Give a brief note on 1x16 decoder with an example.

15. Explain master Slave Flip Flop with a neat diagram.

16. Expound on Shift register and its types with a neat diagram.

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